MIL-M-5793A(ASG)
30 JUNE 1954

Superseding
MIL-M-5793(USAF)
4 April 1950

MILITARY SPECIFICATION

MICROPHONE M-3A/A

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 This specification covers one type of a boom-suspended, noise cancelling, carbon microphone assembly designated as Microphone M-3A/A, for use in military aircraft at altitudes under 15,000 feet.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

NN-P-515	Plywood, Container Grade
QQ-M-151	Metals; General Specification for Inspection of
ZZ-R-601	Rubber Goods; General Specifications (Methods of Physical Tests and Chemical Analyses)
PPP-B-601	Boxes, Wood, Cleated-Plywood

Military

MIL-A-140	Adhesive, Water-Resistant, Waterproof Barrier-Material
MIL-B-131	Barrier Material; Water Vaporproof, Flexible
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-M-5658	Mildew Proofing of Fabrics, Threads and Cordage; Copper Processes for
MIL-M-5794	Microphone Unit M-6A/UR
MIL-P-116	Preservation, Methods of
JAN-P-100	Packaging and Packing for Overseas Shipment - General Specification
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed
JAN-P-125	Packaging and Packing for Overseas Shipment - Barrier-Materials, waterproof, Flexible
JAN-P-642	Plugs, Telephone

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STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS25070	Microphone Boom, MT-521A/U
MS25071	Bracket, MT-522A/U

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The requirements, including materials and workmanship specified in Specification MIL-E-5400 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.2 Assembly.- Each Microphone M-3A/A shall consist of one MT-521A/U Microphone Boom in accordance with Standard MS25070, one MT-522A/U Microphone Bracket in accordance with Standard MS25071, and one M-6A/UR Microphone Unit in accordance with Specification MIL-M-5794.

3.2.1 Microphone boom MT-521A/U.-

(a) The microphone clip shall be designed to accommodate Microphone Unit M-6A/UR in accordance with Specification MIL-M-5794. The clip shall hold the microphone unit securely enough to prevent the unit from falling out of the clip when Microphone M-3A/A is subjected to the Drop test specified herein. However, the unit shall not be clamped so tightly that it prevents easy removal of the microphone unit from the clip. A small slot shall be provided in one edge of the clip to permit insertion of a small screwdriver to aid in removal of the microphone unit.

(b) The centers of contact springs on the microphone clip shall coincide with the centers of the contacts on Microphone Unit M-6A/UR when the unit is mounted properly in the clip.

(c) The microphone cord shall be clamped securely to the microphone clip without damage to the conductors or cordage jacket. Care shall be taken when soldering the conductors in the clip in order to insure that the lead dress is such that the conductors are not pinched or otherwise damaged when the microphone unit is mounted in the microphone clip.

(d) The cordage guides shall have openings sufficiently large to accept microphone cordages with an outside diameter of 0.187 inch. All edges and corners shall be flared or rounded to prevent damage to the cordage.

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3.2.2 Bracket MT-522A/U.-

(a) The main functions of the bracket are to hold the microphone boom in the normal operating or standby positions and to provide adjustments for these positions and a slide adjustment for the wire-loop mounting of the microphone boom. The slide or the swivel bracket shall be designed to hold the microphone boom in any desired position by spring tension. The force required to move the microphone boom, along its axis, in the swivel bracket shall be between 3 and 5 pounds.

(b) Means shall be provided in the bracket assembly to prevent the detent wheel from falling out of its mounting when the swivel bracket is moved to its extreme positions.

(c) The design of Bracket MT-522A/U shall be such that it may be used on either the right or left sides of the headset and such that the reversal may be made without the aid of tools.

3.3 Design.- The design of Microphone M-3A/A shall be such that it is convenient to adjust for use in the normal operating or standby positions through the full range of standard human head sizes.

3.4 Insulation.- The insulation provided between all respectively insulated parts of the microphone, with the microphone unit removed, shall successfully withstand the application of 110 volts, 60 cps, ac, for a period of 1 minute without flashover or breakdown.

3.5 Swivel and detent action.- The swivel, slide, and detent action of the microphone assembly shall be smooth and free from binding. The torque required to displace the boom from either of the detent positions shall be within the limits of 55 and 100 ounce-inches. The torque required to just start the boom moving from rest from any position other than the detent positions shall be not greater than 55 ounce-inches. All torque measurements shall be made with the plane of rotation horizontal. The detent device shall be positive and shall hold the boom firmly in position without play.

3.5.1 Resistance to salt spray.- There shall be no binding of movable parts or impairment of the mechanical or electrical operation of the boom and bracket after they have been subjected to a salt spray for 100 hours.

3.6 Microphone cord.- The microphone cord shall consist of the specified length of two-conductor tinsel cordage terminated in a Plug PJ-292, and shall include an identification tag shown on Standard MS25070.

3.6.1 Cordage.- The insulation and jacket compounds shall be homogeneous in character, tough, and elastic, and shall be free from blisters, cracks, and other imperfections which would affect the serviceability of the cordage.

3.6.1.1 Jacket.- The jacket shall provide long life under severe service conditions. The tensile strength of the jacket shall be not less than 1,200 psi, and the elongation of the jacket shall be not less than 300 percent in 2 inches. The jacket shall have a minimum thickness of 0.015 inch and shall not adhere or stick to the conductor insulation. The finished cordage shall have a reasonably smooth surface.

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3.6.1.2 Conductor insulation.- The conductor insulation shall be "free-stripping" and shall not have any detrimental properties which would make soldering of the conductors difficult. The tensile strength of the conductor insulation shall be not less than 1,200 psi, and the elongation of the conductor insulation shall be not less than 100 percent in 2 inches.

3.6.1.3 Accelerated aging.-

3.6.1.3.1 Oxygen bomb - Specimens of the jacket and the conductor insulation, after being aged 96 hours in an oxygen bomb, shall not show any signs of cracking or fracture. The tensile strength and elongation shall not be reduced by more than 25 percent.

3.6.1.3.2 Extreme heat.- Specimens of the jacket and conductor insulation, after being aged for 96 hours in an oxygen bomb, shall meet the temperature requirements of this specification. The tensile strength and elongation shall not be reduced by more than 25 percent. The jacket shall not show any signs of blistering, cracking, warping, or other deterioration as a result of this test.

3.6.1.4 Central cores, fillers, binders, etc.- All cotton or other textiles used as a central core, filler, binder, etc, shall be mildewproof-treated in accordance with Specification MIL-C-5658. The impregnating material shall not affect the electrical or physical characteristics of the cordage.

3.6.1.4.1 Stay cord - A central core or filler shall extend throughout the length of the cordage and shall be suitable for use as a strain-relief cord. The central core or filler shall be a low-stretch material with a 20-pound minimum breaking strength.

3.6.1.5 Flammability.- With the finished cord fully extended in a horizontal position in still air, self-sustained combustion of the insulation shall not progress at a rate in excess of 1 inch per minute.

3.6.1.6 Outside diameter.- The cordage shall have a maximum outside diameter of 0.165 inch.

3.6.1.7 Microphone cord guides.- The microphone cord guides shall freely pass a cordage having a maximum diameter of 0.165 inch.

3.6.1.8 Temperature requirements.-

3.6.1.8.1 Operating range.- The cordage shall be flexible and resilient throughout the temperature range of -55° to +85°C.

3.6.1.8.2 Cold bend.- Specimens of the cordage shall show no evidence of cracking or other damage to the jacket or conductor insulation after being subjected to a cold test. The bent specimens shall withstand, while immersed in tap water, a potential of 500 volts rms, 60 cps, applied between each conductor and the water for 1 minute.

3.6.1.9 Flexing.- The cordage shall be capable of withstanding at least 50,000 flexing cycles without showing any evidence of

(a) Damage to the jacket of the cordage
(b) Damage to the insulation of the individual conductors
(c) Electrical discontinuity

3.6.1.10 Electrical requirements.- The finished cordage shall meet the electrical requirements stated below.

3.6.1.10.1 Conductor resistance.- The resistance of an individual conductor of the finished cordage shall not exceed 0.6 ohm per foot at 20°C.

3.6.1.10.2 Dielectric strength.- The finished cordage shall withstand, without breakdown, a potential of [illegible] volts rms, 60 cps, applied for 1 minute, as specified in sec

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3.6.1.10.3 Insulation resistance.- The insulation resistance of the finished cord shall be not less than 500 megohms per 1,000 feet at a temperature of 15.5°C, when tested as described in Section 4.

3.6.2 Plug PJ-292.- Plug PJ-292 shall be in accordance with Specification JAN-P-642.

3.6.3 Cord assembly.- The Plug PJ-292 shall be assembled to the cordage in such a manner that a force of 20 pounds applied between the two items is taken by the central core or filler cords of the cordage, and an anchor point within the plug. The 20-pound pull shall be applied in the direction tending to separate the cordage from the plug, and shall not cause the cordage to slip or pull out of the plug or otherwise damage the conductors. The cable clamp portion of Plug PJ-292 shall grip the cordage in such a manner that a 10-pound pull tending to separate these items will not cause the cordage to slip from the plug. The central core or filler cords or conductors of the cordage shall not be anchored within the plug when testing for this requirement.

3.7 Identification of product.- The microphone and the subassemblies and parts shall be marked for identification in accordance with Standard MIL-STD-130. The marking shall be applied to the cordage by means of a metal wrap-around tag or a suitable plastic sleeve. The marking shall include the following information:

Microphone M-3A/A
1/ Contract or Order No.
Manufacturer's name or trade-mark

1/ Data to be inserted by contractor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.- The inspection and testing of microphones shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests performed on samples representative of the production of the item after the award of contract, to determine that the production meets the requirements of this specification.

(b) Acceptance tests: Acceptance tests are those tests performed on individual lots which have been submitted for acceptance.

4.2 Preproduction tests.-

4.2.1 Sampling instructions.- Ten microphones and 50 feet of cordage used in the microphones, as specified in the contract, will be tested for design approval by the procuring activity, or when so specified in the contract, at the contractor's plant under the supervision of the procuring activity. The preproduction samples shall be in all respects truly representative of the microphones the contractor proposes and expects to furnish in quantity production, and shall be accompanied by adequate identifying information, including the following, where applicable:

Name of manufacturer
Name of contractor and Part No.
Contract or Order No.

4.2.1.1 In case the item or part has been approved previously, the date of approval, the name of the approving activity, and the previous contract or order number shall be included in the above sample identifying information.

4.2.1.2 Plug PJ-292.- The sample requirements for Plug PJ-292 shall be in accordance with Specification JAN-P-642. The two portions of the shell shall comply with the requirements for shells.

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4.2.2 Tests.- The Preproduction tests shall consist of all the tests described under Acceptance tests, the ones specified below, and the Preproduction tests specified in Specification MIL-M-5794. Plug PH-292 shall be tested in accordance with Specification JAN-P-642. The cordage shall be tested in accordance with the following.

4.2.2.1 Tensile strength, elongation, and accelerated aging.- These tests shall be performed in accordance with Specification ZZ-R-601.

4.2.2.2 Flexing test.- Five sample lengths of cordage, each approximately 1-1/2 feet long, shall be clamped and suspended through holes in a rectangular (1/2 by 5/8 inch) metal bar in the manner described below:

(a) The clamping point shall be immediately above the metal bar.
(b) The cordage shall be clamped in such a manner that it will not turn in the hole during the flexing test.
(c) The length of the through holes in the metal bar shall be 5/8 inch. The diameter of the through holes shall not exceed the diameter of the cord by more than 0.010 inch.
(d) The through holes in the metal bar shall incorporate a 1/8-inch radius at the end from which the cord emerges (bending point).
(e) A knot shall be tied near the free end of the cord, and a 1-pound weight shall be hung from the knot.

4.2.2.2.1 The two conductors of each sample length of cordage, a small low current drain 6-volt lamp, and a suitable power source shall be connected in series to indicate electrical continuity. The metal bar, supported horizontally, shall then be rotated about its longitudinal axis back and forth through an angle of 120 degrees (60 degrees each side of vertical) at the rate of approximately 85 cpm.

4.2.2.3 Current capacity.- A current of 6.5 ampere shall be passed through a loop consisting of both conductors of a 3-foot length of cordage, using a proper low-voltage, d-c source. The current shall be applied for a period of 3 hours, after which the cordage shall be examined for detrimental effects.

4.2.2.4 Resistance.- The conductor resistance shall be measured by means of a Leeds and Northrop No. 4286 Kelvin Bridge Ohmmeter, or its equivalent.

4.2.2.5 Dielectric strength.- The finished cordage shall be immersed in tap water at room temperature for 24 hours with the end terminals entirely above the surface of the water. At the end of this period, the specified voltage shall be applied between the two conductors, and between each of the two conductors and the water in which the cordage is immersed.

4.2.2.6 Insulation resistance.- The insulation resistance shall be measured immediately following the Dielectric strength test with a suitable resistance measuring device, using a 500-volt d-c potential. The conductors shall be maintained negative with respect to the ground. The insulation resistance shall be determined 1 minute after voltage application, or sooner if the reading of the resistance measuring device remains steady.

4.2.2.7 Resistance to salt spray.- The booms and brackets shall be subjected to a salt spray in accordance with Specification QQ-M-151. Immediately after removal from the salt spray chamber, they shall be washed and examined.

4.2.2.8 Cold bend.- Specimens of the finished cord, each about 2 feet long, shall be skinned 1 inch at each end to the bare conductor. They shall then be placed in a cold chamber at -60°C (±2°C) with their upper ends attached to a mandrel 1 inch in diameter and with their lower ends weighted. The specimens shall then be conditioned for a period of at least 20 hours. At the end of that period, and while still at -60°C, they shall be wrapped about the mandrel at the rate of one turn in 4 seconds for a sufficient number of turns in order that their center portions are subjected to at least five turns. The specimens shall be unwrapped and rewrapped in the opposite direction in a similar manner. The weights attached to the lower ends of the specimens shall be sufficient to

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keep them taut during the bending operations, in order that close turns of the specimens upon the mandrel will be obtained. The specimens shall then be removed from the mandrel and, without straightening, immersed in tap water not more than 30°C for 1 hour, with 6 inches of each end protruding above the water. They shall then be subjected to the Dielectric test.

4.2.2.9 Drop test.- Microphone M-3A/A shall be subjected to a Drop test consisting of 20 random drops from a height of 5 feet to a concrete floor.

4.3 Acceptance tests.- The Acceptance tests shall consist of Individual tests and Sampling tests.

4.3.1 Individual tests - Each microphone submitted for acceptance under contract shall be subjected to the following tests.

4.3.1.1 Examination of product.- Every part shall be subject to such inspection as the procuring activity deems necessary to determine compliance with the mechanical detail, material, and workmanship requirements of this specification. Such inspection may be of component parts, subassemblies, partial assemblies, or the completed part, as the mechanical simplicity or complexity of the part indicated necessary or desirable.

4.3.1.2 Insulation.- The insulation between all respectively insulated parts of the microphone, with the microphone unit removed, shall be checked.

4.3.1.3 Swivel slide and detent action.- The swivel slide and detent action of the microphone shall be checked.

4.3.1.4 Talk-out test - Each microphone complete with microphone cover shall be mounted on a Headset HS-33-A and given a talk-out test, using the test circuit shown in figure 1 with switches S-1 and S-4 open and the headset connected to the output of Meter M-2 through a Headset Adapter MC-3[illegible]-( ), or equivalent transformer. Each microphone shall perform satisfactorily.

4.3.2 Sampling tests.- One microphone shall be selected at random, from each lot of 100 or fraction thereof on the order, for subjection to the Sampling tests which shall consist of all the tests described herein, and, in addition, the Sampling tests specified in Specifications MIL-M-5794 and JAN-P-642.

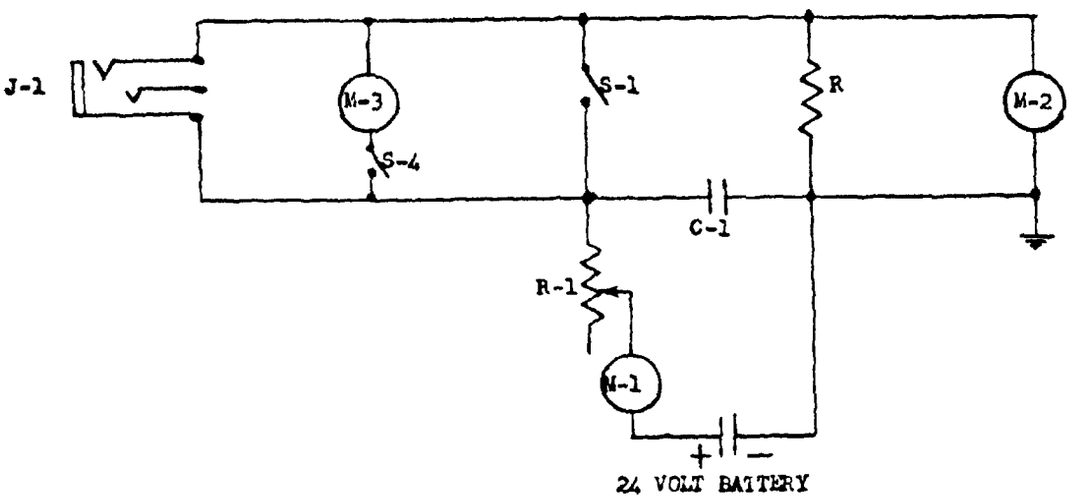
4.3.3 Rejection and retest - When tests are specified on a quantity of microphones that are selected as representative of a certain lot, and one or more of this number fails to meet the requirements, additional microphones of the lot represented shall be tested immediately to determine the cause of failure. Individual performance tests shall not be interrupted unless the defect is of such a nature that it will seriously affect the performance or safe use of the microphone.

5. PREPARATION FOR DELIVERY

5.1 Application.- The requirements of Section 5 apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging.- The microphone shall be either commercially packaged or unit-pack packaged, as specified by the procuring activity.

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J-1 MICROPHONE JACK, 3-CONTACT
S-1 SWITCH, SPST
C-1 CAPACITOR, 100 MFD, 50 VOLT, ELECTROLYTIC
R-1 VARIABLE RESISTOR, 400 OHMS
R RESISTOR, 100 OHM
M-1 MILLIAMMETER, 0-200 MA, DC
M-2 ELECTRONIC VOLTMETER, BALLANTINE MODEL 300, OR EQUAL
M-3 VOLTMETER, 0-10 VOLTS, DC, 1,000 OHMS PER VOLT MINIMUM
S-4 SWITCH, PUSH BUTTON, SINGLE CONTACT

FIGURE 1. Test circuit

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5.2.1 Commercial packaging.- No packaging beyond that normally provided under best commercial practice will be required.

5.2.2 Unit packaging.- Each microphone shall be unit-packaged in accordance with Specification MIL-P-116, method II, carton-barrier-carton. Barrier materials shall be in accordance with Specification MIL-B-131. Cushioning materials shall be in accordance with Specification JAN-P-100, except that excelsior will not be acceptable.

5.3 Packing.- Each component shall be packed for either domestic or overseas shipment, as specified by the procuring activity. Exterior shipping containers, insofar as possible, shall be uniform in size and shape, shall contain an identical number of like items only, and shall be designed to enclose the contents in a compact manner.

5.3.1 Domestic shipment.- Domestic-type exterior shipping containers shall conform to the requirements of Consolidated Freight Classification Rules in effect at the time of shipment and shall be designed to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery. Where fiberboard is used, it shall have a minimum Mullen Test of 275 pounds.

5.3.2 Overseas shipment.- Overseas-type exterior shipping containers shall be cleated plywood boxes in accordance with Specification PPP-B-601. Plywood, if used, shall be in accordance with Specification NN-P-71b, type II, class 2. As an alternative, nailed wood boxes in accordance with Specification JAN-P-106, style 2 or 4, may be used. In either event, the box shall be lined with a waterproof case liner of material in accordance with Specification JAN-P-125, sealed with material in accordance with Specification MIL-A-14C.

5.4 Marking of shipments.- Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows: Microphone M-3A/A, Specification MIL-M-5793A, Manufacturer's Part No. *, Federal Stock No. * (if no FSN available, leave space therefor).

*Applicable information to be entered by the contractor.

6. NOTES

6.1 Intended use.- The microphone is intended for use in military aircraft at altitudes under 15,000 feet.

6.2 Ordering data.- Invitations for bids, contracts, and purchase orders should state the conditions for the following.

6.2.1 Preproduction test.- It is expected that the contract or purchase order will specify that 10 microphones and 6 feet of cordage will be required as First Articles (or preproduction samples) and that these First Articles (or preproduction samples) will be subjected to the Preproduction tests to determine compliance with the requirements of the specification. The invitation for bids and the contract should specify the point of inspection for these tests.

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Custodians:
Navy - Bureau of Aeronautics
Air Force

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE PLATE N 15419

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